

**DEMO UNIVERSITY**  
**END SEMESTER EXAMINATION**  
**DATA STRUCTURES AND ALGORITHMS**

Demo Question Paper

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|-----------------------------------------------|--------------------------|
| <b>Course:</b> Data Structures and Algorithms | <b>Time:</b> 2 Hours     |
| <b>Date:</b> 24 May 2025                      | <b>Maximum Marks:</b> 50 |

**Instructions**

- Answer all questions.
- Each question carries 5 marks.
- Draw neat and properly labelled diagrams wherever required.
- Assume suitable data wherever necessary and clearly state your assumptions.

**Section A — Descriptive Questions**

- Q1.** Define Stack. Write its applications. **[5 Marks]**
- Q2.** Explain the working of Queue with a diagram. **[5 Marks]**
- Q3.** What is the time complexity of Binary Search? Explain. **[5 Marks]**
- Q4.** Write short notes on Tree Traversal. **[5 Marks]**
- Q5.** Given a linked list, explain how insertion at the beginning is performed. **[5 Marks]**
- Q6.** Explain Linked List and write its types with a diagram. **[5 Marks]**
- Q7.** What is Hashing? Explain with an example. **[5 Marks]**
- Q8.** Explain the working of Stack using an example. **[5 Marks]**
- Q9.** What is Recursion? Write a recursive function to calculate factorial of a number. **[5 Marks]**
- Q10.** Differentiate between Array and Linked List. **[5 Marks]**

— End of Question Paper —